

CONTACT
INFORMATION

225B Kildee Hall
806 Stange Rd
Iowa State University
Ames, IA, 50011, USA

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WWW: yebigithub.github.io

RESEARCH
INTERESTS

I am a Postdoctoral Research Associate from Dr. Juan Steibel's group in the Department of Animal Science at Iowa State University. I earned my Ph.D. under the mentorship of Dr. Gota Morota at Virginia Polytechnic Institute and State University. My research interests focus on incorporating artificial intelligence, computer vision, statistics, phenomics, and genetics to study animal and plant sciences.

CURRENT
RESEARCH
PROJECTS**Precision Livestock Farming**

- Develop and implement models for automatic behavioral phenotyping that require minimal amounts of training data and that can be validated in broader contexts.
- Segmentation and tracking of group-housed pigs using foundation models.
- Computer vision for farrowing detection and piglet tracking via zero-shot models.
- Dairy cattle competitive behavior classification around GreenFeed and gas emission estimation.
- Early identification of fall-behind weaned pigs with supervised learning and large vision models.

EDUCATION

Virginia Polytechnic Institute and State University, Blacksburg, Virginia USA

Ph.D., Animal and Poultry Science, August 2021 - December 2024

- Thesis: "Digital phenotyping and genomic prediction using machine and deep learning in animals and plants"
- Advisor: Dr. Gota Morota
- Committee: Drs. Mark Hanigan, Ismini Lourentzou, and Nicholas Santantonio

Graduate Certificate in Data Analytics, January 2024 [[URL](#)]

- Completed 15 credits related to advanced data science courses from the Departments of Computer Science, Electrical and Computer Engineering, and Statistics.

University of California Davis, Davis, California USA

M.S., Animal Biology, September 2021

- Thesis: "Longitudinal Analysis of CD4 and CD8 T Cell Receptor Repertoires Associated with Newcastle Disease Virus Infection in Layer Birds"
- Advisor: Dr. Huaijun Zhou
- Committee: Drs. Charles L. Bevins and Rodrigo A Gallardo
- Available at [UC Davis Libraries](#).

Chinese Academy of Agricultural Sciences, Beijing, Beijing China

M.S., Animal Nutrition and Feed Science, July 2017

- Thesis: "Effects of Dietary Threonine Level on Traits of Peking Ducks from Hatch to 21 Days"
- Advisor: Dr. Benhai Xiong
- Committees: Drs. Shuisheng Hou, Jilan Chen, Taozhen Jiang, and Yu Chen

Shandong Agricultural University, Taian, Shandong China

B.S., Animal Science, July 2014

- Thesis: “Effect of Rumen Fluid Osmotic Pressure on Absorption of VFAs in Rumen Epithelium of Sheep”
- Advisors: Drs. Yunliang Jiang and Zhonghua Wang

PROFESSIONAL
POSITIONS

Department of Animal Science
Iowa State University, Ames, Iowa, USA

- Postdoctoral Research Associate
- Advisor: Dr. Juan Steibel

01/2025 - current

WORK
EXPERIENCES

School of Animal Sciences
Virginia Polytechnic Institute and State University, Blacksburg, Virginia, USA

- Graduate Research Assistant

09/2021 - 12/2024

Precision Animal Health Group
Zoetis, Inc., Kalamazoo, Michigan USA

- Precision Animal Health Data Science Intern
- Advisor: Dr. Di Liang

05/2024 - 08/2024

Department of Animal Biology
University of California Davis, Davis, California USA

- Graduate Research Assistant

08/2019 - 08/2021

PREPRINT

2026

1. **Ye Bi**, Bimala Acharya, David Rosero, and Juan Steibel. Automated Segmentation and Tracking of Group Housed Pigs Using Foundation Models. arXiv (2026). doi: [10.48550/arXiv.2604.03426](https://doi.org/10.48550/arXiv.2604.03426)

PEER-REVIEWED
RESEARCH JOURNAL
ARTICLES

2026

18. Anna Wolc, Usamah Kabuye, Tricia Veldhuizen, Jaedyn Condon, **Ye Bi**, Kaylee Rowland. Cell Phone Image Analysis as a Proxy for Laboratory Egg Quality Measurements. In press. doi: [10.1016/j.psj.2026.1074706](https://doi.org/10.1016/j.psj.2026.1074706)
17. **Ye Bi**, Yijian Huang, Haipeng Yu, and Gota Morota. Impact of Trait Measurement Error on Quantitative Genetic Analysis of Computer Vision-Derived Traits. Genes 17, no. 5 (2026): 506. doi: [10.3390/genes17050506](https://doi.org/10.3390/genes17050506)

- 2025
16. **Ye Bi**, Yijian Huang, Jianhua Xuan, and Gota Morota. Industry-scale prediction of video-derived pig body weight using efficient convolutional neural networks and vision transformers. *Biosystems Engineering* 257 (2025): 104243. doi: [10.1016/j.biosystemseng.2025.104243](https://doi.org/10.1016/j.biosystemseng.2025.104243).
 15. Mingsi Liao, Gota Morota, **Ye Bi**, and Rebecca Cockrum. Predicting dairy calf body weight from depth images using deep learning (YOLOv8) and threshold segmentation with cross-validation and longitudinal analysis. *Animals* 15, no. 6 (2025): 868. doi: [10.3390/ani15060868](https://doi.org/10.3390/ani15060868).
 14. **Ye Bi**, Harkamal Walia, Toshihiro Obata, and Gota Morota. Genomic prediction of metabolic content in rice grain in response to warmer night conditions. *Crop Science*, 65(1), p.e21435. doi: [10.1002/csc2.21435](https://doi.org/10.1002/csc2.21435).
- 2024
13. Salem, Mohamed, Rafet Al-Tobasei, Ali Ali, Liqi An, Ying Wang, Xuechen Bai, **Ye Bi**, and Huaijun Zhou. Functional annotation of regulatory elements in rainbow trout uncovers roles of the epigenome in genetic selection and genome evolution. *GigaScience* 13 (2024): giae092. doi: [10.1093/gigascience/giae092](https://doi.org/10.1093/gigascience/giae092)
 12. Sabag, Idan, **Ye Bi**, Maitreya Mohan Sahoo, Ittai Herrmann, Gota Morota, and Zvi Peleg. Leveraging genomics and temporal high-throughput phenotyping to enhance association mapping and yield prediction in sesame. *The Plant Genome* (2024): e20481. doi: [10.1002/tpg2.20481](https://doi.org/10.1002/tpg2.20481)
 11. Kenan Burak Aydin, **Ye Bi**, Luiz F. Brito, Zafer Ulutaş, and Gota Morota. Review of sheep breeding and genetic research in Türkiye. *Frontiers in Genetics*. **15**:1308113. doi: [10.3389/fgene.2024.1308113](https://doi.org/10.3389/fgene.2024.1308113)
- 2023
10. **Ye Bi**, Leticia M. Campos, Jin Wang, Haipeng Yu, Mark D. Hanigan, and Gota Morota. Depth video data-enabled predictions of longitudinal dairy cow body weight using thresholding and Mask R-CNN algorithms. *Smart Agricultural Technology*. **6**:100352. doi: [10.1016/j.atech.2023.100352](https://doi.org/10.1016/j.atech.2023.100352)
 9. Zhangyuan Pan, Ying Wang, Mingshan Wang, Yuzhe Wang, Xiaoning Zhu, Shenwen Gu, Conghao Zhong, Liqi An, Mingzhu Shan, Joana Damas, Michelle M. Halstead, Dailu Guan, Nares Trakooljul, Klaus Wimmers, **Ye Bi**, Shang Wu, Mary E. Delany, Xuechen Bai, Hans H. Cheng, Congjiao Sun, Ning Yang, Xiaoxiang Hu, Harris A. Lewin, Lingzhao Fang, and Huaijun Zhou. An atlas of regulatory elements in chicken: A resource for chicken genetics and genomics. *Science Advance*. **9**:eade1204(2023). doi: [10.1126/sciadv.ade1204](https://doi.org/10.1126/sciadv.ade1204)
 8. **Ye Bi**, Rafael Massahiro Yassue, Puneet Paul, Balpreet Kaur Dhatt, Jaspreet Sandhu, Thi Phuc Do, Harkamal Walia, Toshihiro Obata, and Gota Morota. 2023. Evaluating metabolic and genomic data for predicting grain traits under high night temperature stress in rice. *G3: Genes, Genomes, Genetics*. doi: [10.1093/g3journal/jkad052](https://doi.org/10.1093/g3journal/jkad052)
 7. Idan Sabag, **Ye Bi**, Zvi Peleg, and Gota Morota. 2023. Multi-environment analysis enhances genomic prediction accuracy of agronomic traits in sesame. *Frontiers in Genetics*. **14**:1108416. doi: [10.3389/fgene.2023.1108416](https://doi.org/10.3389/fgene.2023.1108416)
- 2022
6. Robert Kadlec, Sam Indest, Kayla Castro, Shayan Waqar, Leticia M Campos, Sabrina T Amorim, **Ye Bi**, Mark D Hanigan, and Gota Morota. 2022. Automated acquisition of top-view dairy cow depth image data using an RGB-D sensor camera. *Translational Animal Science*. **6**:1-6. doi: [10.1093/tas/txac163](https://doi.org/10.1093/tas/txac163)
- 2017

5. **Ye Bi**, Xuemei Nan, Shanshan Zheng, Linshu Jiang, Benhai Xiong. Effects of dietary threonine and immune stress on growth performance, carcass trait, serum immune parameters, and intestinal muc2 and NF-kb gene expression in Pekin ducks from hatch to 21 days. *Poultry Science*, 2017, 97(1): 177-187. doi: [10.3382/ps/pex283](https://doi.org/10.3382/ps/pex283)
 4. **Ye Bi**, Hairui Xin, Xiaohua Pan, Benhai Xiong. Effects of dietary threonine level on growth performance, carcass traits, immune function and serum hormone of Peking ducklings. *Chinese Journal of Animal Nutrition*, 2017, 29(6): 1913-1920. doi: [link](#)
 3. **Ye Bi**, Xiaohua Pan, Hairui Xin, Benhai Xiong. Research progress of the influence of threonine on poultry nutrition[J]. *China Animal Husbandry and Veterinary Medicine*, 2017, 44(8): 2326-2332. doi: [link](#)
- 2016
2. Hairui Xin, Xiaohua Pan, Liang Yang, **Ye Bi**, Benhai Xiong. Effects of Light Intensity on Performance, Carcass Performance and Meat Quality of Peking Ducks. *Chinese Journal of Animal Nutrition*, 2016, 28(4): 1076-1083. doi: [link](#)
 1. Hairui Xin, Xiaohua Pan, **Ye Bi**, Benhai Xiong, Linshu Jiang. Effects of Lighting Regimes on Production Performance, Carcass Performance and Anti-Oxidant Capacity of the Blood in Peking Ducks. *Journal of Integrative Agriculture*, 2016, 49(23): 4638-4645. doi: [link](#)
- CONTRIBUTED PRESENTATIONS
- 2026
7. **Ye Bi**. Data structure to facilitate behavior phenotyping of pigs using foundation vision-language models. World Congress on Genetics Applied to Livestock Production 2026. Madison, Wisconsin. July 12-17. 2026.
- 2025
6. **Ye Bi**. Foundation model-based approach for video segmentation and tracking of group-housed pigs. 2025 National Swine Improvement Federation (NSIF) Meeting. Kansas City, Missouri. October 14-16. 2025.
 5. **Ye Bi**. Automated segmentation and tracking of group housed pigs using zero-shot vision-language tools. 2025 ASAS-CSAS Annual Meeting. Hollywood, Florida. July 6-10. 2025.
 4. **Ye Bi**. The impact of trait measurement error on genetic analysis of image-based phenotypes in pigs. Genomic Selection and Genome-Wide Association Studies session. The Plant and Animal Genome Conference (PAG 32). Town and Country Resort and Conference Center, San Diego, CA, USA. January 10-15. 2025.
- 2024
3. **Ye Bi**. Industry-scale prediction of video-derived pig body weight using efficient convolutional neural networks and vision transformers. 2024 ASAS-CSAS-WSASAS Annual Meeting. Calgary TELUS Convention Centre, Calgary, Alberta, Canada. July 21-25, 2024.
 2. **Ye Bi**. The impact of trait measurement error on quantitative genetic analysis. 2024 ASAS-CSAS-WSASAS Annual Meeting. Calgary TELUS Convention Centre, Calgary, Alberta, Canada. July 21-25, 2024.
- 2023
1. **Ye Bi**. Depth video data-enabled predictions of longitudinal dairy cow body weight using thresholding and Mask R-CNN algorithms. 2023 ASAS-CSAS-WSASAS Annual Meeting. Albuquerque, New Mexico. July 16-20.

INTRAMURAL SEMINARS

- 2025
6. **Ye Bi.** Automated tracking of group-housed pigs using zero-shot vision-language tools. S25 Animal Science Departmental Seminars (Lightning Talk). Ames, IA. May 7th, 2025
 5. **Ye Bi.** Harnessing computer vision and deep learning for genetic analysis of animals. Animal Breeding and Genetics Seminar. Department of Animal Science. Iowa State University, Ames, IA. January 24th. 2025.
- 2024
4. **Ye Bi.** Digital phenotyping and genomic prediction using machine and deep learning in animals and plants. Exit Seminar. School of Animal Sciences. Virginia Polytechnic Institute and State University, Blacksburg, VA. September 18.
- 2023
3. **Ye Bi.** Depth video data-enabled predictions of longitudinal dairy cow body weight using thresholding and Mask R-CNN algorithms. School of Animal Sciences Research Day. Virginia Polytechnic Institute and State University, Blacksburg, VA. May 16.
- 2022
2. **Ye Bi.** Utility of Metabolites and Single-Nucleotide Polymorphisms for Classification of High Night Temperature Stress Conditions and Prediction of Grain Size Related Traits in Rice. Virginia Polytechnic Institute and State University, Blacksburg, VA, Sep 09.
 1. **Ye Bi.** Evaluating Dairy Cow Body Condition Scores Using Automated Computer Vision Systems. Department of Animal and Poultry Sciences Research Day. Virginia Polytechnic Institute and State University, Blacksburg, VA. May 19.

INVITED PRESENTATIONS

- 2026
5. **Ye Bi.** Artificial Intelligence for Livestock Phenomics: From Real-Time Monitoring to Genetic Selection. The University of Maryland, College Park, MA, USA. July 10-11. 2026.
- 2025
4. **Ye Bi.** The impact of trait measurement error on genetic analysis of image-based phenotypes in pigs. Genomic Selection and Genome-Wide Association Studies session. The Plant and Animal Genome Conference (PAG 32). Town and Country Resort and Conference Center, San Diego, CA, USA. January 10-15. 2025.
- 2024
3. **Ye Bi.** Enhancing pig body weight estimation and genetic analysis using computer vision and deep learning. ANSC Seminar. Department of Animal Science. The Pennsylvania State University, University Park, PA. September 5. 2024.
- 2023
2. **Ye Bi.** Animal data science applied to digital data. Special Seminar. Smithfield Premium Genetics. Rose Hill, NC. August 14. 2023.
- 2022
1. **Ye Bi.** Evaluating dairy cow body condition scores using automated computer vision systems. Department of Animal Science. Shandong Agricultural University. Online. July 22. 2022.

POSTERS

- 2026 5. **Ye Bi**. Data structure to facilitate behavior phenotyping of pigs using foundation vision-language models. World Congress on Genetics Applied to Livestock Production 2026. Madison, Wisconsin. July 12-17. 2026.
- 2025 4. **Ye Bi**. Foundation model-based approach for video segmentation and tracking of group-housed pigs. 2025 National Swine Improvement Federation (NSIF) Meeting. Kansas City, Missouri. October 14-16. 2025.
- 2023 3. **Ye Bi**. Depth video data enabled prediction of dairy cow body weight. Virginia Tech Center for Advanced Innovation in Agriculture (CAIA) Big Event. Poster Presentation. March 16. 2023.
- 2022 2. **Ye Bi**. Development of Automated Computer Vision Systems for Evaluating Dairy Cow Body Weight and Body Condition Scores. ASAS-CSAS 2022 annual meeting, Oklahoma City, Oklahoma. Poster Presentation. June 26-30. 2022.
1. **Ye Bi**. Evaluating Dairy Cow Body Condition Scores Using Automated Computer Vision Systems. Virginia Tech Center for Advanced Innovation in Agriculture (CAIA) Big Event. Poster Presentation. March 28. 2022.
- PEER-REVIEWED
CONFERENCE
PROCEEDINGS
- 2026 12. **Ye Bi**, Bimala Acharya, David Rosero, Mina Shumaly, and Juan Steibel. Data structure to facilitate behavior phenotyping of pigs using foundation vision-language models. In World Congress on Genetics Applied to Livestock Production Digital Archive, vol. 2026, no. 1. Iowa State University Digital Press, 2026. doi: [link](#)
11. Isaac Berg, **Ye Bi**, Jeremy Perez, David Rosero, Gustavo Silva, and Juan Steibel. Analysis of visual features extracted from farrowing sows. In World Congress on Genetics Applied to Livestock Production Digital Archive, vol. 2026, no. 1. Iowa State University Digital Press, 2026. doi: [link](#)
10. Bimala Acharya, **Ye Bi**, Anna Johnson, David Rosero, Mina Shumaly, and Juan Steibel. Modeling Behavioral Patterns in Group-Housed Pigs Using Continuous Trajectories. In World Congress on Genetics Applied to Livestock Production Digital Archive, vol. 2026, no. 1. Iowa State University Digital Press, 2026. doi: [link](#)
- 2025 9. **Ye Bi**, Mina Shumaly, Bimala Acharya, David S. Rosero, and Juan P. Steibel. 42 Automated segmentation and tracking of group housed pigs using zero-shot vision-language tools. Journal of Animal Science 103, no. Supplement_3 (2025): 18-19. doi: [link](#)
8. Bharadwaj, Sanjana, Pranjal Ranjan, Yingqi Pei, Jay Yim, **Ye Bi**, Dong Ha, Gota Morota, and Sook Shin. Non-Invasive Pig Weight Prediction using SAM-Enhanced Deep Learning and Depth Imaging. In SoutheastCon 2025, pp. 794-799. IEEE, 2025. doi: [link](#)
- 2024 7. **Ye Bi**, Jianhua Xuan, Yijian Huang, and Gota Morota. 465 Comparative analysis of semantic segmentation and deep regression models with supervised pre-training for accurate prediction of pig body weight from video data: Insights from industry-scale datasets. Journal of Animal Science 102, no. Supplement_3 (2024): 414-415. doi: [link](#)

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| | 6. Ye Bi , Yijian Huang, and Gota Morota. 88 The impact of trait measurement error on quantitative genetic analysis. <i>Journal of Animal Science</i> 102, no. Supplement_3 (2024): 25-26. doi: link |
| 2023 | 5. Ye Bi , Leticia Campos, Jin Wang, Haipeng Yu, Mark Hanigan, and Gota Morota. 151 Depth Video Data-Enabled Predictions of Longitudinal Dairy Cow Body Weight Using Thresholding and Mask R-Cnn Algorithms. <i>Journal of Animal Science</i> 101, no. Supplement_3 (2023): 44-45. doi: link |
| 2022 | 4. Zhangyuan Pan, Ying Wang, Liqi An, Ye Bi , Dailu Guan, Mary E. Delany, Hans H. Cheng, Huaijun Zhou. Functional annotations of regulatory elements in the chicken genome. <i>Proceedings of 12th World Congress on Genetics Applied to Livestock Production (WCGALP)</i> . doi: link |
| | 3. Ye Bi , Robert Kadlec, Kayla Castro, Sam Indest, Sabrina Amorim, Gota Morota. Development of Automated Computer Vision Systems for Evaluating Dairy cow Body Weight and Body Condition Scores. <i>ASAS-CSAS 2022 annual meeting</i> , Oklahoma City, Oklahoma. June 26-30. doi: link |
| 2016 | 2. Ye Bi , Hairui Xin, Benhai Xiong. Effects of dietary threonine level on traits of Peking ducks from hatch to 21 days. <i>The 12th National Conference about Animal Nutrition of Animal Nutrition Branch of Chinese Association of Animal Science and Veterinary Medicine</i> . doi: link |
| | 1. Hairui Xin, Ye Bi , Benhai Xiong. Effects of Lighting Regimes on Blood Calcium Phosphate Level and Anti-Oxidant Capacity of the Blood in Peking Ducks. <i>The 12th National Conference about Animal Nutrition of Animal Nutrition Branch of Chinese Association of Animal Science and Veterinary Medicine</i> . doi: link |

EDITORIAL ACTIVITIES

Ad Hoc Reviewer

- Number of manuscripts reviewed per journal: *Journal of Animal Science* (10), *Computers and Electronics in Agriculture* (2), *Smart Agriculture Technology* (7), *Scientia Agricola* (3), *Artificial Intelligence in Agriculture* (1), *Animals* (5), *AgriEngineering* (1), *Agriculture*(1), *Sensors* (1), *Italian Journal of Animal Science*(1), *Brazilian Journal of Animal Science* (1), *Preventive Veterinary Medicine* (2).

TEACHING

Iowa State University, Ames, Iowa, USA

Guest Lectures

- ANS 4550/5550 Digital Technologies in Animal Production Systems **Spring/2026**
Introduction to Computer Vision for Animal Scientists

Virginia Polytechnic Institute and State University, Blacksburg, Virginia, USA

Guest Lectures

- ALS 3104 Animal Breeding and Genetics **Spring/2024**
Use of Genetic Markers in Animal Breeding

- ALS 3104 Animal Breeding and Genetics **Spring/2023**
Statistical Concepts in Quantitative Genetics

Teaching Assistant

- ALS 3104 Animal Breeding and Genetics **Spring/2024**
- ALS 3104 Animal Breeding and Genetics **Spring/2023**
- ALS 3104 Animal Breeding and Genetics **Spring/2022**

University of California Davis, Davis, California, USA

Teaching Assistant

- ANS150 Animal Health and Disease **Spring/2020**
- ABI102 Animal Biochemistry and Metabolism **Fall/2019**

**HONOR AND
AWARDS**

- | | |
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| 2024 | <ul style="list-style-type: none"> • Agricultural Genome to Phenome Initiative (AG2PI) Student Conference Travel Award (\$1500), U.S. Department of Agriculture (USDA), National Institute of Food and Agriculture (NIFA). • CAIA Graduate Student Travel Awards program (\$1000), Virginia Tech. |
| 2023 | <ul style="list-style-type: none"> • The Summer 2023 Cycle Travel Fund Program (\$300), Graduate and Professional Student Senate, Virginia Tech. |
| 2022 | <ul style="list-style-type: none"> • Modern Programming in Genome to Phenome Scholarship (\$1500), University of California, Davis, CA, August 1-5. • 27th Summer Institute in Statistical Genetics (SISG) Scholarship, University of Washington, Seattle, WA, July 18-27. |
| 2020 | <ul style="list-style-type: none"> • UC Davis Henry A. Jastro Graduate Research Awards • UC Davis Animal Biology Graduate Program Fellowship |
| 2016 | <ul style="list-style-type: none"> • Academic Scholarship of Chinese Academy of Agricultural Sciences |
| 2015 | <ul style="list-style-type: none"> • Academic Scholarship of Chinese Academy of Agricultural Sciences • Course Excellence Award of Chinese Academy of Agricultural Sciences |
| 2014 | <ul style="list-style-type: none"> • Academic Scholarship of Chinese Academy of Agricultural Sciences • Outstanding Graduate of Shandong Agricultural University |
| 2013 | <ul style="list-style-type: none"> • Excellent Student Scholarship of Shandong Agricultural University • Science and Technology Innovation Scholarship of Shandong Agricultural University |

2012	• Excellent Student Scholarship of Shandong Agricultural University
2011	• Excellent Student Scholarship of Shandong Agricultural University
ACTIVITY	• Virginia Tech Center for Advanced Innovation in Agriculture(CAIA) Graduate Student Affiliate Group Member October/2021 - December/2024 • The 10th National Congress of Animal Nutrition Branch of Chinese Association of Animal Science and Veterinary Medicine (CAAV) and the 12th Animal Nutrition Symposium member October/2016 - September/2017 • The XXV World's Poultry Congress member September/2016 - August/2017 • The Branch of Animal Information, Chinese Association of Animal Science and Veterinary Medicine the 10th Symposium member July/2015 - June/2016
COURSES TAKEN	Virginia Polytechnic Institute and State University • ECE 6554 Advanced Computer Vision • CS 5824 Advanced Machine Learning • ECE 6524 Deep Learning • ECE 5554 Computer Vision • CS 5525 Data Analytics I • STAT 5364G Advanced Statistical Genomics • ALS 5224 Introduction to Genomic Data Science • HORT 5304 Advanced Plant Genetics and Breeding University of California Davis • STA 200B Introduction to Mathematical Statistics I • GGG 201A Advanced Genetic Analysis • STA 200A Introduction to Probability Theory • PMI 270 Advanced Immunology • PMI 126 Fundamentals of Immunology • STA 106 Applied Statistical Methods: Analysis of Variance • PLS 120 Applied Statistics in Agricultural Sciences
ADDITIONAL TRAINING	• Modern Programming in Genome to Phenome, University of California, Davis, CA, August 1-5, 2022.

- 27th Summer Institute in Statistical Genetics (SISG), University of Washington, Seattle, WA, July 18-27, 2022.
- UIUC Spring Workshop: Applied Quantitative Genetics for Plant Breeders, University of Illinois Urbana-Champaign, Urbana, IL, June 1-3, 2022.

COMPUTER SKILLS

- Statistics/Numerical computational tools: R, SAS
- Programming languages: Python, MATLAB, Julia
- Content-description languages: $\text{\LaTeX}$
- Operating system: Linux and Mac OS X
- Computer clusters: Slurm workload manager

REFERENCES References and additional information available upon request.